



Understanding the influences of In-situ annealing and substrate vibration on the charge carrier dynamics of ultrasonic spray-coated polymer solar cell

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ABSTRACT

Detailed understanding of the various influences of deposition conditions on the structure–property relationship for spray-coated polymer films is crucial for their scalable device applications. In the present study, the influences of in-situ substrate temperature and acoustic substrate vibration on the charge carrier dynamics of poly(3-hexylthiophene) and [6,6]-phenyl-C71-butyric acid methyl ester (P3HT:PC₇₁-BM) based ultrasonic spray-coated polymer solar cells have been investigated thoroughly by employing Impedance spectroscopy, Mott–Schottky analysis, Urbach energy analysis, and trap-state density estimations. The device prepared under the influences of in-situ substrate temperature and acoustic substrate vibration shows more than three times enhancement in PCE (3.24%) compared to that of the reference one (0.9%). A correlation between charge transport behaviour and deposition conditions has been identified for the devices. The surface roughness and rigid droplet boundaries were found to set major performance limitations. The overall resistance of the devices was found to get decreased by 70% whilst the global charge carrier mobility was found to get increased from 6.09×10^{-5} to $9.43 \times 10^{-4} \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ with the simultaneous application of substrate temperature and acoustic vibration, forming uniform and homogeneous films with much reduced surface roughness and droplet boundaries compared to the untreated reference devices. Systematic variation in the trap and defect-state densities were also observed. The trap-state density reduced from 5.03×10^{15} to $2.71 \times 10^{15} \text{ cm}^{-3}$ after the combined treatment of in-situ annealing and substrate vibration. Urbach energy was found to be 218.4 meV for the untreated active layer, which reduced to 177.2 meV for the active layer treated with in-situ-annealing and acoustic substrate vibration. The superior electrical properties achieved by optimizing the active layer morphology using different spray deposition conditions led to around four times enhancement in device efficiency.

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1 Introduction

Third-generation photovoltaic technologies such as dye-sensitized, quantum dot, polymer, and perovskite solar cells have been developed to achieve high performance with low production cost [1–3]. Amongst these technologies, polymer solar cells (PSCs) have gained immense research attention as they are easily compatible with large area manufacturing processes [4, 5]. Moreover, PSCs have owned advantages like lightweight, mechanical flexibility, inexpensive, and solution-based manufacturing. However, to follow the footprint of Si solar cell to its commercialization, PSCs must achieve high efficiency and open-air device stability in the same order [6–9]. For this purpose, the researchers have explored various strategies such as synthesis of new conjugated polymers with promising properties, development of novel device architectures, and optimization of the nanoscale morphology of photoactive layers [10–15]. Nowadays, power conversion efficiency (PCE) more than 18% for single-junction and 19.6% for tandem PSCs along with device lifetime over four months have been reported [16–18]. These studies have been generally carried out in research laboratories by employing thin film fabrication methods like spin coating, dip coating, and drop casting. Generally, such lab scale fabrication techniques are not compatible with the targeted large-scale production of PSCs. To implement the large area Roll-to-Roll (R2R) and Layer-to-Layer (L2L) production of PSCs, large-scale deposition methods like inkjet printing, spray coating, slot die coating, and screen printing have been generally utilized [19, 20]. Amongst these, the spray coating technique has been widely accepted as a promising method for scalable deposition due to its unique advantages such as R2R and L2L compatibility, flexibility of patterning, low wastage, and high throughput [21, 22].

High performance devices have been achieved by exploring the various approaches of spray coating techniques [21, 23–27]. Recently, Zhang et al. reported a PCE of 8.75% in the fullerene-based devices fabricated via conventional two step spray coating [28]. Cheng et al. employed the piecewise spray deposition to fabricate a device based on non-fullerene acceptor and achieved a PCE of 12.29% [29]. In addition, the researchers have also utilized spray deposition techniques to develop semi-transparent

devices and deposit active layer directly on to a polyester cotton fabric [30, 31]. The performance of these devices strongly depends on the spray deposition parameters and interfacial properties of photoactive layer due to their uneven surface morphology and high possibility of charge recombination/leakage site formation [32, 33]. The electrical response and the charge carrier transport of these devices have been generally investigated in a macroscopic level by evaluating the current density–voltage (J–V) characteristics. Although this approach is beneficial in determining the overall photovoltaic parameters and the charge conduction mechanisms to a certain extent, a detailed study on the charge carrier dynamics in the microscopic level is still essential to clearly understand the factors influencing the performance of devices.

Our recent studies [34, 35] revealed that the in-situ annealing of substrate during the spray deposition of active layer provide the devices with the performance comparable to those of spin coated ones. The devices fabricated at an in-situ annealing temperature of 80 °C provided the maximum efficiency. However, the efficiency of these devices was limited to a certain value due to the presence of droplet boundaries, which act as charge trapping sites for photo-generated charge carriers. Further improvement in device performance was achieved by better controlling on the nanoscale morphology of active layer by exciting the substrate with acoustic vibrations. A significant reduction in droplet boundaries was observed by the simultaneous application of in-situ annealing and acoustic vibration, which led to four times improvement in device efficiency. In the present work, detailed electrical measurements such as impedance spectroscopy, capacitance–voltage (C–V) analysis, defects and trap-state measurements and photocurrent analysis were carried out to investigate the charge carrier dynamics of the devices. This enabled us to explain the defect formations and the performance enhancement mechanism of the spray-coated devices fabricated at various deposition conditions, including quantitative characteristics, as well as to describe how the electronic structure of the film and interfaces plays critical role in the carrier transport and device efficiencies.

2 Experimental details

Regioregular poly(3-hexylthiophene) (P3HT) and [6, 6]-phenyl-C71-butyric acid methyl ester (PC₇₁BM) were purchased from 1-material. Poly (3,4-ethylene-dioxythiophene):poly(styrene sulfonate) (PEDOT:PSS) (1.3 wt% dispersion in H₂O, conductive grade), and Lithium fluoride (LiF) were obtained from Sigma Aldrich. All the materials were used as received without any further purification. PSCs with the structure of ITO/PEDOT:PSS/P3HT:PC₇₁BM/LiF/Al were fabricated. Prior to the fabrication, the ITO substrates were cleaned thoroughly in deionized water, soap solution, acetone, and isopropanol in sequence using an ultrasonic bath followed by UV-ozone treatment. The treated substrates were spin coated with filtered PEDOT:PSS at 3500 rpm for 60 s and were baked at 150 °C for 15 min in air. Initially, a solution of P3HT:PC₇₁BM (1:1 wt ratio) of concentration 3 mg ml⁻¹ in chlorobenzene was prepared and kept overnight for stirring at 45 °C. Then, the solution was ultrasonic spray deposited on PEDOT:PSS coated ITO substrates at an air pressure of 20 psi and a flow rate of 1.5 ml min⁻¹ under various conditions as reported in our previous manuscripts [34, 35]. Finally, 1 nm LiF and 100 nm Al layers were thermally evaporated through a shadow mask. Current density–voltage (J–V) characteristics of the devices were measured using a Keithley 2401 source meter with AAA solar simulator with AM1.5G solar spectrum under stimulated radiation condition of 76.2 mW cm⁻². The impedance and C–V measurements were carried out using NF corporation-ZM2376 LCR meter. UV–Vis–NIR absorption spectrophotometer (Perkin Elmer Lambda-1050) was used to investigate the optical characteristics. The thickness of all the films was measured using Dektak XT stylus profilometer. The optical microscope images were taken from Olympus BX53M.

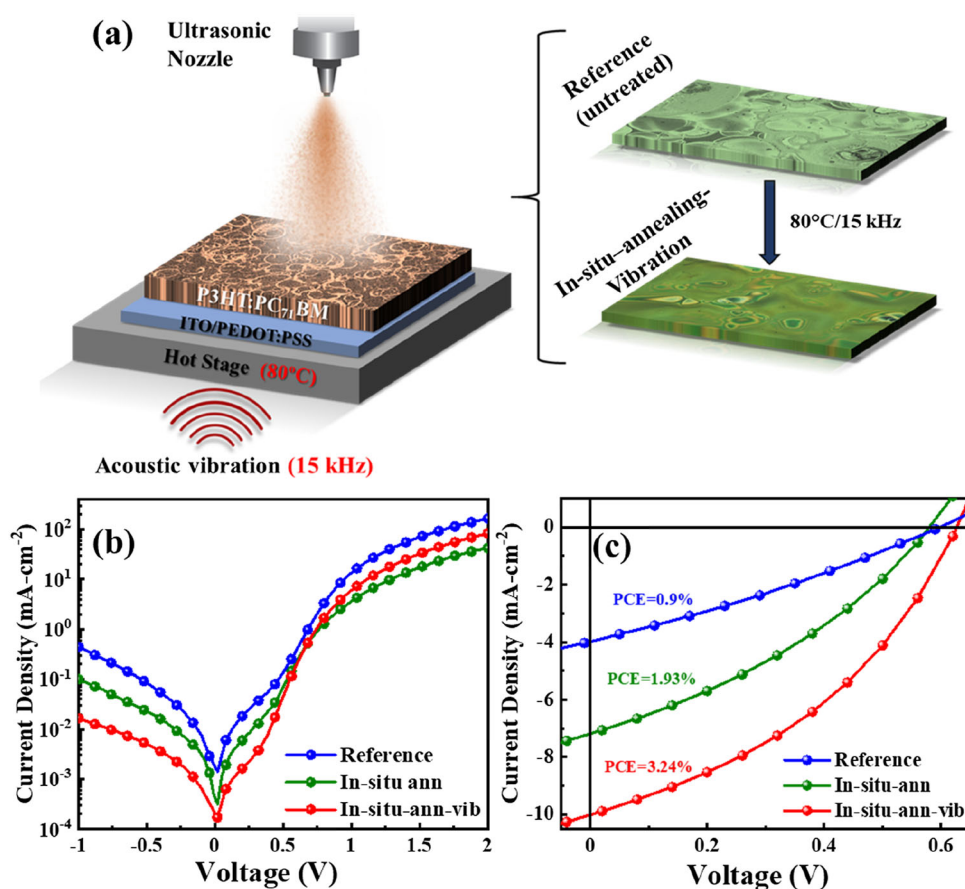
3 Results and discussion

The schematic diagram of the ultrasonic spray deposition set up and the variation in film formation under different deposition conditions are shown in Fig. 1a. As reported in our earlier studies, significant reduction in film roughness and rigid droplet boundaries were observed with the introduction of in-situ substrate annealing (80 °C) and acoustic

substrate vibration (15 kHz) compared to the as prepared reference films [34, 35]. The P3HT:PC₇₁BM active layer was deposited under three different previously optimized conditions of ultrasonic spray deposition; (a) device with no treatment on substrate named as reference device, (b) device with in-situ annealing of 80 °C on substrate, named as in-situ-ann device, and (c) device with in-situ annealing and 15 kHz acoustic vibration on substrate, referred to as in-situ-ann-vib device. The corresponding dark J–V characteristics of the devices are shown in Fig. 1b which reveal that the leakage current in reference device is higher compared to other two conditions. The rectification ratio obtained at ± 1 V for the device deposited with in-situ-annealing-vibration condition was found to be 4.5×10^2 , which is one order of magnitude higher compared to reference device with rectification ratio of 4×10^1 . This clearly indicates that initially the reference films were having large number of leakage paths whilst those leakage paths got significantly reduced with better droplet coalescence under the influence of in-situ annealing and substrate vibration, leading to more uniform and homogeneous films (Fig. S1). The J–V characteristics of the devices under AM 1.5G solar radiation are shown in Fig. 1c and the device performance parameters are summarized in Table S1. The reference devices exhibited inferior performances with average short-circuit current density (J_{sc}) of 3.98 mA cm⁻², open-circuit voltage (V_{oc}) of 0.59 V, and Fill Factor (FF) of 29.10%, leading to PCE of 0.90%. As evident in Fig. 1c, with the application of in-situ annealing, temperature assisted crystallization of P3HT leads to increment in J_{sc} by more than 88% compared to reference device, resulting in PCE of 1.93% and FF of 34.30%. Further performance improvement was observed with the simultaneous application of acoustic vibration during the spray processing.

The average J_{sc} reaches to 10 mA cm⁻² whilst V_{oc} of 0.63 V were observed for the devices fabricated under in-situ-annealing-vibration condition, leading to the maximum PCE and FF of 3.24% and 38.90%, respectively. The higher V_{oc} for in-situ-annealing-vibration condition compared to reference and in-situ annealing condition can be attributed to smaller reverse saturation current density (J_0), as open-circuit voltage has inverse relation with J_0 [36]. Such significant enhancement in device performance indicates strong influences of solvent evaporation (assisted by

Fig. 1 **a** Schematic for ultrasonic spray deposition process, **b** Dark J-V characteristics, and **c** Light (AM 1.5G) J-V characteristics of spray-coated devices at various deposition conditions



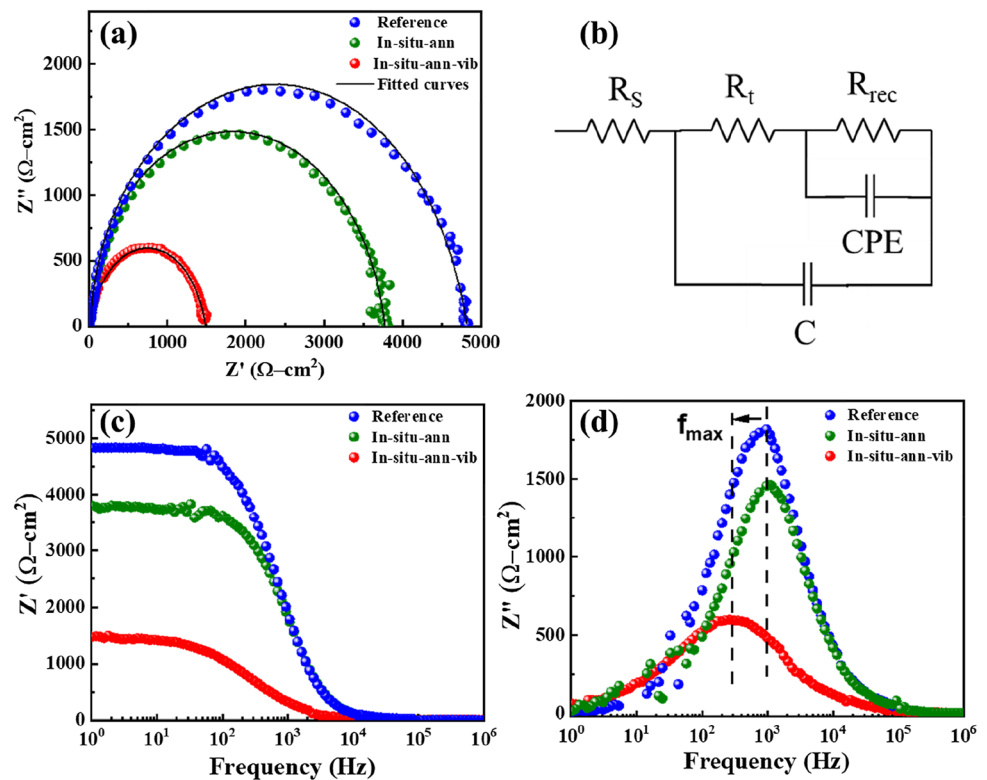
in-situ annealing) and reduction in droplet boundaries (assisted by substrate vibration) on the charge carrier dynamics of ultrasonic spray-coated polymer solar cell. Therefore, understanding the structure–property relationship of the spray-coated films in terms of their electronic structure in the bulk as well as at the interfaces is crucial to identify the limiting factors for the devices in question.

To understand the influence of in-situ annealing and substrate vibration on the charge carrier dynamics of the devices, the impedance spectroscopy was performed. Detailed analysis of the charge carrier dynamics in the bulk and the interfacial regions, and the charge distribution (polarization) in the active layer of the devices, was carried out [37, 38]. Figure 2a shows the Nyquist plot for the corresponding devices at open circuit and dark conditions, in the frequency range from 1 Hz to 1 MHz with 40 mV of AC signal. The semicircle features of the plots are clearly visible where the diameter of the semicircle signifies the total internal transport resistance in the device which comprises of electrode,

bulk, and interface charge transfer resistances. For the reference devices, the overall internal resistance was found to be around $4.9 \text{ k}\Omega \text{ cm}^2$. As evident from Fig. 2a, a significant reduction in the diameter was observed as the in-situ annealing and in-situ-annealing-vibration conditions were introduced with average value of $3.7 \text{ k}\Omega \text{ cm}^2$ and $1.5 \text{ k}\Omega \text{ cm}^2$, respectively. Film roughness and the rigid droplet boundaries often act as the trap centres for the photogenerated charge carriers [34]. Since all the devices were prepared with identical ultrasonic spray deposition parameters except for substrate condition, it can be attributed that the observed reduction in the internal transport resistances of the devices is likely due to the modification in the active layer morphology and active layer/cathode interfaces.

The equivalent circuit modelling was carried out to further understand the effect of spray deposition conditions in detail. However, due to slight asymmetry in the Nyquist plots, a simple electrical circuit model consisting of a single $R||C$ with a series resistance was found to be insufficient to fit these

Fig. 2 Impedance analysis **a** Nyquist plot of the devices deposited in different condition, **b** equivalent circuit model; Garcia-series model, **c** resistance vs frequency, and **d** reactance vs frequency of the devices deposited under different condition



plots accurately. Therefore, **Garcia-Belmonte series model**, as shown in Fig. 2b, containing more than one $R||C$ element is adopted here which gives good fit to these impedance responses [39–42]. The resistance R_s represents the overall series resistance of the devices whilst R_t corresponds to the charge transport resistance and R_{rec} is the recombination resistance or charge transfer resistance of the devices. The global capacitance, i.e. the dielectric behaviour of the device is represented by C . A constant phase element (CPE, represented by an exponent n) is added in the model to incorporate the equivalent capacitive effects of randomly distributed P3HT:PC₇₁BM clusters in the blend film. For the ideal condition the n should be 1 [43, 44]. **Effective carrier lifetime, transit time (diffusion time), and the global mobility were further investigated using the fitting parameters from the equivalent circuit model and the following formula [39]:**

Effective capacitance,

$$C_\mu = \sqrt[n]{PR_{rec}}/R_{rec} \quad (1)$$

Effective lifetime,

$$\tau_n = R_{rec}C_\mu \quad (2)$$

Diffusion time,

$$\tau_d = R_t C_\mu \quad (3)$$

Global mobility,

$$\mu_n = eL^2/\tau_d k_B T \quad (4)$$

where P is CPE fitting parameter, L is the thickness of the active layer (200 nm), k_B is Boltzman constant, T is temperature.

All the fitted components of the equivalent circuit model and the estimated parameters from the expressions (1) – (4) are summarized in Table 1. Interestingly, a drastic fall in both R_s and R_t (R_s ~ from 27.06 to 5.30 $\Omega \text{ cm}^2$, R_t ~ from 2.59 to 0.05 $\text{k}\Omega \text{ cm}^2$) was observed for the in-situ-annealed-vibration samples compared to the reference devices, indicating reduced contact and charge transport resistance in the treated devices. **The recombination resistance (R_{rec}), which accounts for the recombination of injected charge carriers, seems to be reducing as well with modified deposition conditions.** It is worth noting that the lower value of R_t than R_{rec} is responsible for improved transport and reduced recombination of charge carriers in OSC devices [45]. As evident from Table 1, the ratio of R_t and R_{rec} was found to be more than 1 for reference device.

Table 1 The fitted components of equivalent circuit, lifetime, and mobility of P3HT:PC₇₁BM devices deposited under different conditions

Devices	Series resistance, R_s ($\Omega \text{ cm}^2$)	Transport resistance, R_t ($\Omega \text{ cm}^2$)	Recombination resistance, R_{rec} ($\Omega \text{ cm}^2$)	n	Chemical capacitance, C_μ (nF cm^{-2})	Effective lifetime, τ_n (μs)	Diffusion time, τ_d (μs)	Global mobility, μ_n ($\text{cm}^2 \text{ V}^{-1} \text{ s}^{-1}$)
Reference	27.06	2598.90	2302.20	0.75	97.5	224	253	6.09×10^{-5}
In-situ-ann	25.99	1354.20	2244.9	0.79	178.90	401	242	6.38×10^{-5}
In-situ-ann-vib	5.30	49.70	1420.4	0.85	329.25	468	16	9.43×10^{-4}

However, it decreased for in-situ annealed and in-situ-annealed vibration devices to 0.6 and 0.035, respectively, explaining the increased current density and reduced leakage current. These results are in good agreement with the results obtained from the J-V characteristics (Fig. 1b).

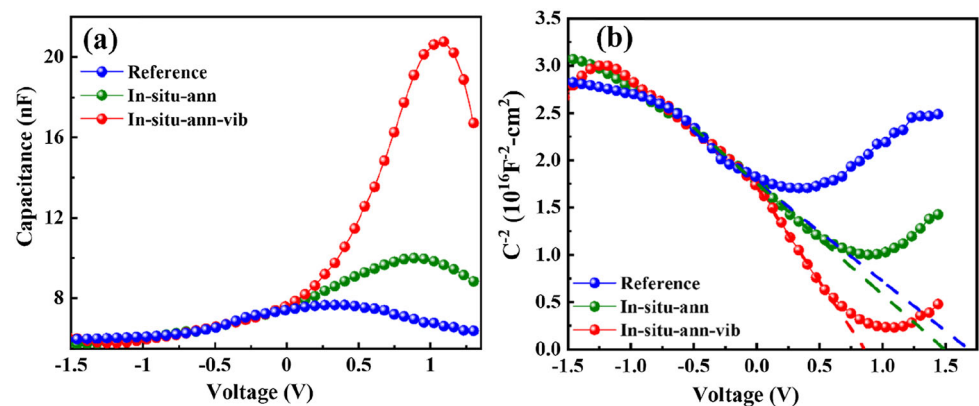
The value of exponent n was also found increasing systematically from 0.75 for the reference device to 0.79 and 0.85 for the in-situ annealing and in-situ-annealed-vibration condition devices, respectively, which further confirm improved phase aggregation and interfacial properties for the bulk heterojunction (BHJ) films fabricated under the combined effects of substrate annealing and acoustic vibrations. The in-situ-annealed-vibration devices exhibited prolonged carrier lifetime ($\tau_n \sim 468 \mu\text{s}$) compared to the in-situ-annealed ($\tau_n \sim 401 \mu\text{s}$) and reference devices ($\tau_n \sim 224 \mu\text{s}$). Longer carrier lifetime leads to efficient charge collection, resulting in enhanced PCE for the in-situ-annealed and in-situ-annealed vibration device. Similarly, lowest diffusion time (τ_d) was observed for the in-situ-annealed-vibration devices (16 μs) compared to the reference devices (253 μs), indicating that less time is needed for the electrons to diffuse out of active layer to the electrode [44]. Systematic improvement in the global charge carrier mobility (μ_n) was also observed with each substrate condition. More than one order increment in μ_n , from 6.09×10^{-5} to $9.43 \times 10^{-4} \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ was observed for the in-situ-annealing vibration devices, compared to the reference device, which is consistent with the observed enhancement in J_{sc} and FF for the corresponding devices.

The frequency dependency of the resistive and reactive component of the impedance spectra are shown in Fig. 2c and d. The resistance of all the devices decreases with increasing frequency and

becomes independent at high frequency, as shown in Fig. 2c. However, this decrease is occurring at lower frequency in case of in-situ-annealing-vibration condition, showing the dominance of capacitance shunting with resistance compared to other two conditions [46]. From the Fig. 2d, it is evident that the maximum reactance value has decreased from 1.8 $\text{k}\Omega \text{ cm}^2$ for reference device to 0.6 $\text{k}\Omega \text{ cm}^2$ for in-situ-annealing-vibration device along with a shift in characteristics frequency (f_{max}) towards lower frequency. The shift in the value of f_{max} has a direct correlation with an increase in characteristic relaxation time for in-situ-annealing-vibration condition, as observed in Table 1. This finding is in accordance with light J-V characteristics of devices, as larger values of relaxation time lead to enhancement in PCE and overall device performance [45].

To understand the effects of the different deposition conditions over the internal built-in potential and carrier distribution of the devices, C-V measurements were carried out under dark condition at low operating frequency (100 Hz). Figure 3a shows the C-V characteristics of spray deposited devices under three substrate conditions. Nearly same capacitance was observed for all the devices at zero bias condition, which indicates that the initial depletion conditions were comparable. However, with applied bias, the capacitance starts increasing and reaches a peak value at a forward voltage. Beyond this the capacitance starts decreasing again due to the injection of charge carriers from the electrodes into the active layer. Interestingly the rate of increment and decrement of capacitance was found to be higher for the in-situ-annealing-vibration condition, suggesting faster polarization of the active layer under applied bias. Such devices are often favourable for efficient exciton dissociation [47, 48].

Fig. 3 **a** C–V characteristics and **b** Mott–Schottky analysis for ultrasonic spray deposited devices at various substrate conditions, the dashed lines represent the linear fitting



The corresponding Mott–Schottky (MS) analysis are shown in Fig. 3b. Previous literature indicates the p-type behaviour of P3HT:PC₇₁BM devices due to the formation of acceptor type impurities via structural defects and air exposure [39, 41]. The acceptor impurity concentrations (N_A) were estimated for the devices by the following relationship:

$$C^{-2} = 2(V_{FB} - V)/eA^2\epsilon_0\epsilon_rN_A \quad (5)$$

where V_{FB} is the Flat-band voltage, N_A is the acceptor impurity concentration, e is the elementary charge of electron, ϵ_r is dielectric constant ($\epsilon_r = 3.5$), ϵ_0 is the vacuum permittivity, and A corresponds to active area of the device ($A = 0.04 \text{ cm}^2$). The values calculated by investigating the slope of C^{-2} – V curve shown in Fig. 3b are given in Table 2. In conventional BHJ device, V_{FB} is defined as the amount of band bending at BHJ/cathode interface and the quantity of applied bias for flattening the energy band. As shown in Fig. 3b, the response of spray-coated devices follows Eq. (5) (MS behaviour), as a straight line was observed for a wide range of bias voltage. This yields V_{FB} of 1.68 V at reference condition, $V_{FB} = 1.48 \text{ V}$ for in-situ annealing, and $V_{FB} = 0.84 \text{ V}$ for in-situ-annealing-vibration condition. The higher value of V_{FB} for spray-coated reference devices in comparison to the in-situ-annealed and in-situ-annealed-vibration

devices, is attributed to rough BHJ surface leading to uneven interface and formation of surface defect states. Such morphological defect states are responsible for unintentional acceptor doping. This unintentional impurity doping in the active layer due to atmospheric oxygen enhances hole concentration and immobile superoxide anions. Which results in an increase of the density of traps for electrons and formation of space charge region at the active layer/metal interface, hindering the charge carrier extraction [9, 49, 50]. The doping concentration follows the similar trend as V_{FB} , and its value decreases from $2.40 \times 10^{18} \text{ cm}^{-3}$ for the reference device to $1.26 \times 10^{18} \text{ cm}^{-3}$ for the in-situ-annealed-vibration devices.

The structural and morphological changes caused by the in-situ annealing and substrate vibration should directly influence the localized defect-state formation in the active layer. Such defect states often cause rapid charge trapping, limiting the cell efficiency and device stability. To further investigate the influence of deposition conditions on the defect-state formation in the active layer, quantitative analysis was carried out using Urbach energy calculations for P3HT:PC₇₁BM spray deposited films at different substrate conditions. Figure 4a shows the absorption coefficient vs photon energy plot for the corresponding films where the Urbach energy (E_u), is estimated from the slope of the exponential tail of the plots by following linear equation [51, 52]:

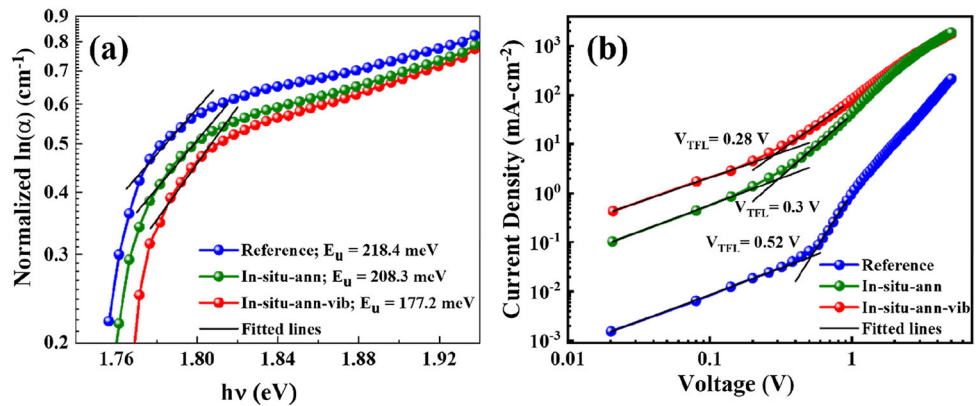
$$\ln\alpha = \ln\alpha_0 + (h\nu/E_u) \quad (6)$$

where h is Planck's constant, E_u is the Urbach energy, and α_0 is a fitting parameter. It is clear from Fig. 4a that the tail is getting gradually sharper with the application of in-situ substrate temperature and substrate vibration. The Urbach energy (E_u) was

Table 2 Parameters obtained from Mott–Schottky analysis of spray deposited devices

Devices	V_{FB} (V)	N_A (cm^{-3})
Reference	1.68	2.40×10^{18}
In-situ-ann	1.48	2.16×10^{18}
In-situ-ann-vib	0.84	1.26×10^{18}

Fig. 4 **a** Urbach energy analysis for spray deposited P3HT:PC₇₁BM film under different conditions, and **b** Dark J-V characteristics of electron only devices revealing trap-filled-limit voltage (V_{TFL})



found to be 218.4 meV for the reference devices whilst it reduces to 208.3 meV and 177.2 meV for the in-situ-annealed and in-situ-annealed-vibration films, respectively. The 19% drop in Urbach energy for the in-situ-annealed-vibration films compared to the reference films indicates significant reduction in the localized defect states with modified deposition conditions. This is consistent with the observed film morphologies where uniform and homogenous films with reduced rigid droplet boundaries were obtained under the influence of in-situ annealing and acoustic substrate vibrations. In consequence, improved charge transport and increase in J_{sc} and FF are obvious and expected.

As discussed in MS analysis section, the acceptor impurity doping leads to electron trap-state formation. Therefore, electron only devices with the active layers being deposited under similar spray conditions like the actual devices were prepared to calculate the electron trap densities. Figure 4b shows their corresponding dark J-V characteristics in log-log scale. Initially, for all the samples the Ohmic region (slope = 1.1) is visible where the current increases linearly with applied bias. However, with further increase in voltage, non-linear current increment was observed, suggesting that the traps are getting filled with injected charge carriers. Once all the traps are filled the current starts getting saturated. Interestingly, for the reference devices such current saturation is not prominent. The trap-state density (N_t) for the devices were calculated by measuring their trap filled limit voltages (V_{TFL}) [53, 54]:

$$N_t = (2\epsilon_0\epsilon_r V_{TFL})/eL^2 \quad (7)$$

where e is the elementary charge of electron, L stands for the thickness of the active layer (200 nm), ϵ_r dielectric constant ($\epsilon_r = 3.5$), and ϵ_0 is the vacuum permittivity. The N_t for the reference device was found to be $5.03 \times 10^{15} \text{ cm}^{-3}$, whilst it significantly decreased to $2.90 \times 10^{15} \text{ cm}^{-3}$ for in-situ annealed and $2.71 \times 10^{15} \text{ cm}^{-3}$ for in-situ-annealed-vibration devices, due to reduction in their V_{TFL} values from 0.52 V for reference device to 0.30 V and 0.28 V for in-situ annealed and insitu-annealed-vibration devices, respectively.

To understand the various effects of the deposition conditions over the carrier generation and exciton dissociation mechanism, the dependency of photocurrent density ($J_{PH} = J_{Light} - J_{Dark}$) on effective voltage ($V_{eff} = V_0 - V$) for the devices were investigated, where V_0 is the voltage at which there is no photocurrent, as shown in Fig. S2 [55, 56]. At moderate voltage the photocurrent increases linearly, primarily dominated by the diffusion of charge carriers. However, with increasing electric field, the carrier drift dominates the photocurrent and gets saturated at higher V_{eff} . The maximum generation rate (G_{max}) and exciton dissociation probability are estimated from the saturated photocurrent density (J_{PH-sat}). The maximum generation rate ($G_{max} = J_{PH-sat}/eL$) for the in-situ-annealed-vibration devices ($4.28 \times 10^{21} \text{ cm}^{-3} \text{ s}^{-1}$) was found to be 2.5 times higher compared to the reference devices ($1.77 \times 10^{21} \text{ cm}^{-3} \text{ s}^{-1}$). This is in accordance with our earlier observations. On the other hand, the exciton dissociation probability, which is the ratio of the photocurrent density and saturated photocurrent density at short-circuit condition, were found to be 68%, 72%, and 75% for reference, in-situ annealing,

and in-situ-annealed-vibration devices, respectively. This clearly suggests that both carrier generation and exciton dissociation properties of the devices are significantly improved under in-situ annealing and substrate vibrations [55, 57]. The results demonstrate the importance of the fundamental understanding of the correlation between the morphology manipulation (reduction of pinholes, surface roughness, and droplet boundaries) and device performances for rational optimization of efficient spray-coated devices.

4 Conclusion

In summary, the performance of PSCs was significantly improved by the simultaneous application of in-situ substrate annealing and acoustic substrate vibration during the spray deposition of P3HT:PC₇₁-BM active layer. A significant reduction in droplet boundaries was observed in the active layer films when these spray deposition conditions were applied. Various measurements and analysis were carried out to investigate the influence of spray deposition conditions on the charge carrier dynamics of devices. The electrical properties of devices were found to be very sensitive to the spray deposition conditions. The devices fabricated under in-situ-annealing-vibration conditions exhibited relatively lower overall resistance $1.5 \text{ k}\Omega \text{ cm}^2$ and higher global charge carrier mobility $9.43 \times 10^{-4} \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ compared to other devices. After the combined treatment of in-situ annealing and acoustic vibration, the charge carrier lifetime increased from 224 μs to 468 μs and its diffusion time decreased from 253 μs to 16 μs indicating the significant reduction in recombination. C–V measurements revealed the faster polarization of in-situ-annealed-vibration treated active layer indicating that it is more favourable for efficient dissociation of photo-generated excitons. Around 19% drop in Urbach energy was observed in the polymer film after treating under in-situ-annealed-vibration condition. Amongst all the devices, the in-situ-annealed-vibration device exhibited the lowest trap-state density value of $2.71 \times 10^{15} \text{ cm}^{-3}$. The present study provides valuable information on the structure–property relationship of spray-coated photoactive layers, which may be useful for the scalable applications of PSCs.

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Author contributions

Conceptualization: SK, SW, and SP; Methodology: SW, SP, and AT; Formal analysis and investigation: SW, SP, AT, RS, and PS; Writing original draft preparation: SW, SP, AT, and SK; Writing review and editing: SK; Funding acquisition: SK, and SW; Supervision: SK.

Data availability

Data available on request from the authors.

Declarations

Conflict of interest The authors declare that there are no conflicts of interest.

Supplementary Information: The online version contains supplementary material available at <http://doi.org/10.1007/s10854-022-08437-w>.

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